THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The longest rod that fits in a room 2 m by 6 m by 9 m has length (in m): 1

(a) 17 **(b)** 12 **(c)** 10 **(d)** 11

2. The legs of a right triangle are in the ratio $9 : 40$ and its hypotenuse is 82 cm. Its perimeter (in cm) is: 1

(a) 180 **(b)** 98 **(c)** 164 **(d)** 182

3. The sides of a rectangle are in the ratio $20 : 21$ and its diagonal is 29 cm. The area of the rectangle (in cm²) is: 1

(a) 29 **(b)** 82 **(c)** 421 **(d)** 420

4. A Pythagorean triplet whose smallest member is 26 is: 1

(a) 26, 168, 170 **(b)** 26, 167, 169 **(c)** 26, 27, 28 **(d)** 26, 169, 171

5. A rhombus has perimeter 356 cm and one diagonal 78 cm. Its area (in cm²) is: 1

(a) 3120 **(b)** 6279 **(c)** 6942 **(d)** 6240

6. A ladder 41 m long, standing in a street, rests against a house on one side when its top reaches 40 m. Without moving its foot it is turned to rest against the opposite house, reaching 9 m. The width of the street (in m) is: 1

(a) 51 **(b)** 41 **(c)** 49 **(d)** 50

7. In a rectangle ABCD, P is a point inside with PA = 4, PB = 6, PC = 12. Then PD^2 equals: 1

(a) 196 **(b)** 164 **(c)** 124 **(d)** -8

8. The longest rod that fits in a room 6 m by 6 m by 7 m has length (in m): 1

(a) 12 **(b)** 11 **(c)** 10 **(d)** 19

9. A Pythagorean triplet whose smallest member is 31 is: 1

(a) 31, 481, 482 **(b)** 31, 479, 480 **(c)** 31, 32, 33 **(d)** 31, 480, 481

10. A rhombus has perimeter 388 cm and one diagonal 130 cm. Its area (in cm²) is: 1

(a) 4680 **(b)** 9425 **(c)** 9360 **(d)** 12610

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a rectangle ABCD, P is a point inside with PA = 5 cm, PB = 11 cm, PC = 13 cm. Find PD^2. 2

12. The legs of a right triangle are in the ratio $7 : 24$ and the hypotenuse is 75 cm. Find the legs. 2

13. Find a Pythagorean triplet in which one member is 36. 2

14. In a rectangle ABCD, P is a point inside with PA = 10 cm, PB = 9 cm, PC = 14 cm. Find PD^2. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 85 m long, with its foot in a street, reaches a window 84 m high on one side. Without moving its foot it is turned to reach a window 40 m high on the opposite side. Find the width of the street. 3

16. A rhombus has perimeter 388 cm and one diagonal 130 cm. Find the other diagonal and the area. 3

17. Two ships leave a port, one sailing east at 16 km/h and the other south at 30 km/h. How far apart are they after 5 hours? 3

OR

A ladder 73 m long, with its foot in a street, reaches a window 55 m high on one side. Without moving its foot it is turned to reach a window 48 m high on the opposite side. Find the width of the street. 3

18. A rhombus has perimeter 436 cm and one diagonal 120 cm. Find the other diagonal and the area. 3

OR

Two ships leave a port, one sailing east at 24 km/h and the other south at 7 km/h. How far apart are they after 3 hours? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 15 cm, BC = 8 cm, CD = 144 cm and DA = 145 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A room is 6 m long, 6 m wide and 7 m high. Find the length of the longest rod that can be placed in it. Also find the diagonal of its floor. 5

20. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 7 cm, BC = 24 cm, CD = 60 cm and DA = 65 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A room is 3 m long, 4 m wide and 12 m high. Find the length of the longest rod that can be placed in it. Also find the diagonal of its floor. 5

END OF QUESTION PAPER Total: 40 Marks

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